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CART263 Creative Computation II

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Project I Artist Statement

Project URL: <https://philippe-bn.github.io/cart263/projects/project-1/>

This piece is titled *Aquarium*, and it is an interactive web experience. It is a simulation of an aquarium, as the name suggests, in which the viewer can move their mouse around to interact with the fish in the aquarium. *Aquarium* attempts to be just that: a believable, somewhat realistic, pretty, water-filled tank inhabited by my fish. I am aware that my tank is not a tank, that the water is not water and that my fish are not fish: my world is a simulation. It exists entirely in simulacrum.

Simulation is representation, but it is not reality. In Jean Baudrillard's *Simulacra and Simulation*, the social theorist divides simulations into phases:

Whereas representation attempts to absorb simulation by interpreting it as a false representation, simulation envelops the whole edifice of representation itself as a simulacrum.

Such would be the successive phases of the image:

it is the reflection of a profound reality;

it masks and denatures a profound reality;

it masks the absence of a profound reality;

it has no relation to any reality whatsoever: it is its own pure simulacrum. (Baudrillard, p.6)

I believe *Aquarium* lives in the third phase, where my fish do not exist as fish – reality is absent - unless the viewer is told that that is what they are – simulation masks reality. The fish may very well be missiles: the shape of the fish is rough and unrefined; it only very broadly resembles the stereotypical representation of a fish.

Creating with this in mind allows for ample freedom. As Daniel Schiffman writes in the introductory chapter to his book *The Nature of Code*, algorithms can be used to reproduce patterns and behaviours found in nature, they can be used to explore “[...] the unpredictable evolutionary and emergent properties of nature [...]” (Schiffman, 2012). The artistic simulation is free from the constraints of scientific accuracy, yet it relies on the same foundations. Whereas mathematical and

physical concepts must be applied for somewhat faithful representations of natural behaviours, the partial inclusion of these forces and dynamics always leads to a false world, where reality, as truth, is unattainable and absent. This frees me to enact creative choices: I get to select which behaviours I wish to include. This process is teleological: every choice I make is in service of the experience. I make my choices with my viewer's experience in mind: should interacting with *Aquarium* be overwhelming? Calming? Intriguing? In my attempt, *Aquarium* is at times meditative, at times interesting, but really it aims to be a beautiful and sublime experience. Immanuel Kant's classical aesthetics of the Beautiful and the Sublime are explored in the piece. The singular relationship of these aesthetics or feelings with nature¹ is at the core of the initial high-density positive *affect*² brought about by my piece: I want the viewer to feel peace from the colourful bodies and their coordinated, ordered, movements, I want the viewer to feel wonder and awe at the emergent dynamics imitating those found in nature. The initial sense of interest-excitement³ motivates the viewer to interact further with the piece, but this eventually fades away.

The choices I make in the design of this experience reflect on who I am and my experiences. My positionality as an artist is present in the work. I chose to not choose a specific range of colours because I am colourblind. I chose to work with fish because I have been playing fish videos for my boyfriend's cat. I was influenced by my peer, and undergraduate student in Design, Zo Kopyna, by their talk on "worm on a string" in relation to non-representational open-ended play, simulacrum, and affect at the Nostagain Institute Symposium a few weeks ago. These concepts in relationship with design are something I wanted to explore: the idea that non-representational or low-fidelity simulated toys and games lead to open-ended play and a space for imagination and interpretation. To recall Beaudrillard's mask over the absence of reality, this mask can take any form, and this can be the missile just as much as the fish. But within the context of play, and the framing of the simulation as an experience of nature, the experience meets the viewer halfway. In play, wonder is encouraged: the only frame given – a simulation of nature – is for the intended affect of intrigue and excitement at the phenomena of nature. In parallel with the real experience

¹ Many of Kant's descriptions of these aesthetics have to do with feelings experienced in nature (Kant, p.16).

² Here, I use the word "affect" in the context of Affect Theory, as coined by psychologist Silvan S. Tomkins in *Affect, Imagery, Consciousness* (1962), as "the primary motivators of human behavior : we act (learn, think, remember, crave, attach) in relation to fear or surprise, enjoyment or shame" (Tomkins from Frank, p.14).

³ Interest-excitement is an affect explored by Tomkins (Tomkins from Frank, p.52).

of nature, the function of play mirrors the function of magic: it serves as the mediator between the experience and the viewer.

My initial idea was to focus on the interaction between one individual's behaviour and a group behaviour. I wanted to focus on the interplay between a fish following the viewer/user's mouse and a flock of fish that would come in and make the fish lose its agency. The concept was to make a comment on agency in systems and conformity by drawing a clear contrast between both behaviours. This changed in the process because I was fascinated with code from *The Nature of Code*'s Chapter 1: "Vectors", in which a few simple vector calculations allowed for an interesting motion of a ball towards the mouse (Schiffman, 2012). I adapted this code into my fish tank in which I already had fifty instances of my fish simply moving left to right at controlled random speeds and directions (which was already adapted from the Task 4 Squirrels exercise) and it immediately gave me a very satisfying result. I could have stopped there, but I wanted to try to implement the flocking dynamics, and so I built on top of it and started following the code from Chapter 5: "Autonomous Agents" in which Schiffman described how to control the motion of each fish independently with vectors (Schiffman, 2012). As I adapted Schiffman's code into mine, I found that the role of agency was already quite present in the implementation of the dynamics of flocking behaviour. I had not realized how much information each agent possessed in the system, and I became fascinated by this. I moved away from the original narrative experience of the two states and decided to stay on an adapted version of flocking as my main behaviour.

The laborious part was adapting Schiffman's p5 code into my vanilla Javascript code. I did not want to resort to an external vector class or library, so I decided to build my own class, with Schiffman's hints and guesses. I ultimately gathered a few methods in Schiffman's code and attempted to look at p5.vector's GitHub to complete the missing ones, but this was pointless. I could not extract from the GitHub the methods and resorted instead to what I would call educated guessing. I interpolated the methods I had and read the mathematical theorems and formulas I would need to adapt to get the vectorial manipulations I needed.

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